

Lecture 1: EM waves, Double-Slit Experiment, EM Spectra

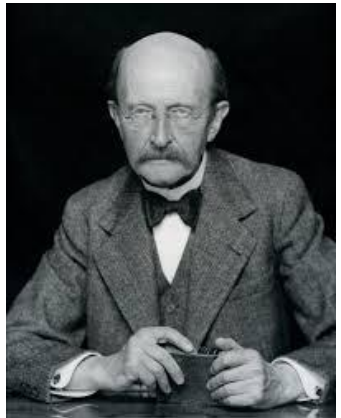
PHY 209

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East West University

Reference Textbooks:

- *Young and Freedman*
- *Kenneth and Krane*

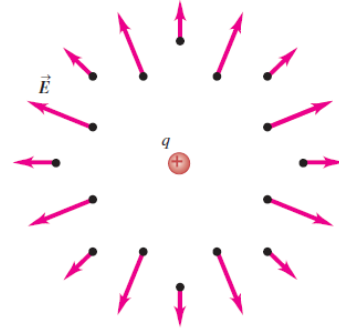


Review of Electric and Magnetic Fields

পুরনো ধারণা রিভিউ করা হয়েছে

- Point charge at rest is a source of *static electric field* ($\vec{E}$)

স্থির চার্জের চারপাশে
Electric Field তৈরি হয়।
like,
Proton
Electron

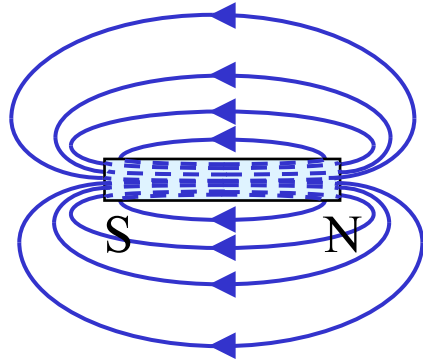
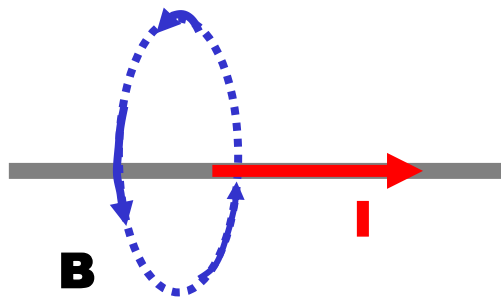


একটি চার্জ q থেকে r দূরে Electric Field:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

চার্জ বেশি $\rightarrow E$ বেশি
দূরত্ব বেশি $\rightarrow E$ কম

- Constant current in a wire or a bar magnet is a source of *static magnetic field* ($\vec{B}$)



$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi}$$

Magnetic Field তৈরি হয়:

Current carrying wire
Magnet.
ফ্যানের মোটর, স্পিকার,
চুম্বক।

- Experiments showed that a **time-varying magnetic field** is a **source of electric field**

- Experiments showed that a **time-varying electric field** is a **source of magnetic field**

এই দুইটি একে অপরকে তৈরি করতে থাকে। এটাই EM Wave-এর ভিত্তি।

Maxwell's Triumph

- Maxwell unified electricity and magnetism into **electromagnetism**, governed by Maxwell's 4 equations

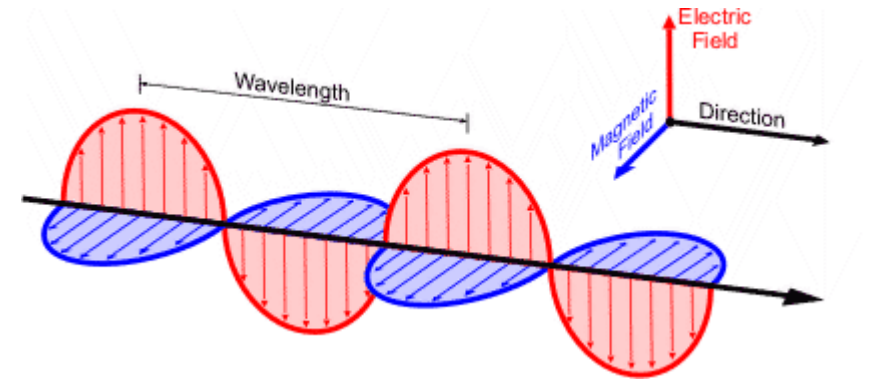
$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \quad \text{Gauss's Law}$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \text{No magnetic monopole}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad \text{Faraday-Lenz Law}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{encl} \quad \text{Ampere's Law}$$

- Thus, accelerating charge or time-varying current produces an EM wave.



- In vacuum, he showed that the speed of EM waves is

আলো আসলে Electromagnetic Wave

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 2.998 \times 10^8 \text{ m/s}$$

আলোর বেগ!

$\mu_0 \equiv$ magnetic permeability in free space

$\epsilon_0 \equiv$ electric permittivity in free space

Electromagnetic (EM) Waves

- EM waves are usually considered to be **plane waves**: the wave fronts are planes
- Mathematically we represent such waves by the following expressions:

Electric Field: $\vec{E} = \vec{E}_0 \sin(kz - \omega t)$ $\vec{B} = \vec{B}_0 \sin(kz - \omega t)$:Magnetic Field

for waves travelling in the z-direction

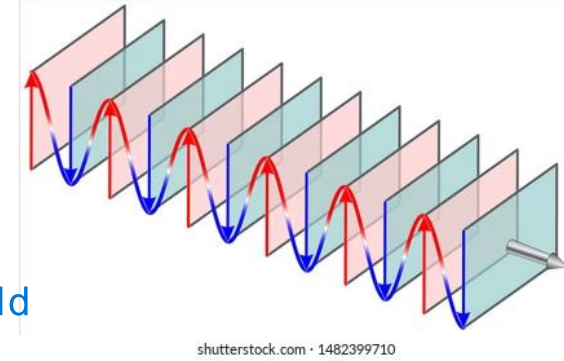
Here: $\vec{E}_0 \equiv$ E-field polarization $\vec{B}_0 \equiv$ B-field polarization

$$k = \frac{2\pi}{\lambda} \quad (\text{wave number}) \quad \omega = 2\pi f \quad (\text{angular frequency})$$

We know: $c = f\lambda \Rightarrow c = \left(\frac{\omega}{2\pi}\right) \left(\frac{2\pi}{k}\right) \Rightarrow c = \frac{\omega}{k}$

Frequency বাড়লে Wavelength কমবে।

- Once the direction of travel and polarization of $\vec{E}_0$ is determined, direction and polarization of $\vec{B}_0$ is automatic since E and B fields must be **mutually orthogonal** (perpendicular)



E_0 -Electric Field-এর Maximum Value

B_0 -Magnetic Field-এর Maximum Value

λ - Wavelength এক তরঙ্গ থেকে পরের তরঙ্গ পর্যন্ত দূরত্ব।

f - Frequency প্রতি সেকেন্ডে কয়টি তরঙ্গ তৈরি হচ্ছে।

k - Wave Number

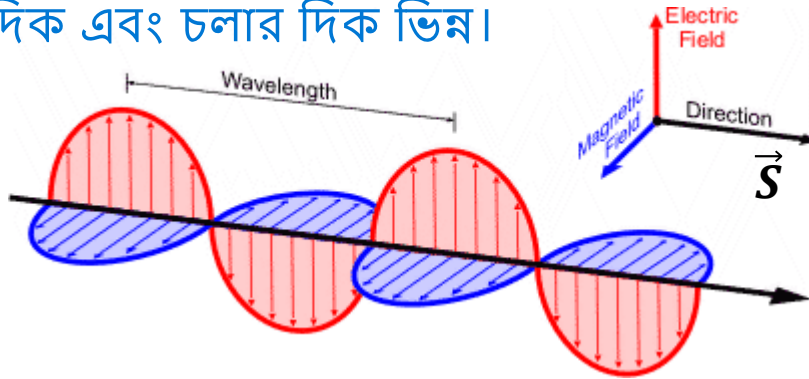
ω - Angular Frequency

Properties of EM radiation

Electric Field এবং Magnetic Field পরস্পর লম্ব।

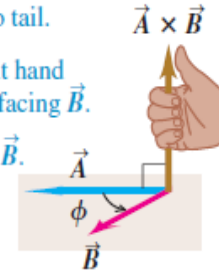
- EM waves are *transverse waves* – direction of wave propagation orthogonal to both the $\vec{E}$ and $\vec{B}$ fields

কম্পনের দিক এবং চলার দিক ভিন্ন।



$\vec{E} \times \vec{B}$ = direction of propagation (Right hand Rule)

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
- ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
- ③ Curl fingers toward $\vec{B}$.
- ④ Thumb points in direction of $\vec{A} \times \vec{B}$.



Right Hand Rule ব্যবহার করে দিক বের করা হয়।

- The ratio of the magnitude of $\vec{E}$ to the magnitude of $\vec{B}$ is the speed of light: $E_0 = cB_0$
- EM waves need no material medium to propagate – otherwise light from the stars or the Sun would never reach us from space

EM Wave চলার জন্য মাধ্যম লাগে না।

যদি মাধ্যম লাগত:

সূর্যের আলো
তারার আলো

পৃথিবীতে আসত না।

কারণ মহাকাশে
শূন্যস্থান।

- An EM wave transmits energy from one place to another, and the energy flux is specified by

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

Poynting Vector

EM Wave শক্তি বহন করে।

এই শক্তির প্রবাহকে বলে:

$$[\vec{S}] = \frac{\text{Power}}{\text{Area}} \rightarrow \text{S. I. unit} = W/m^2$$

Calculating power of EM wave

এখানে Intensity বের করা হয়েছে

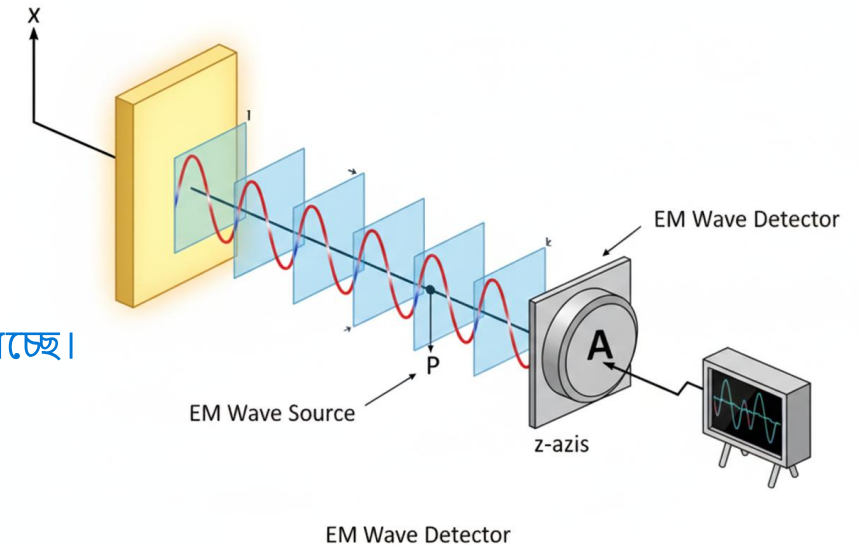
Consider the following thought experiment: we place an EM wave detector at some point on the z-axis. The receiver is oriented in a way such that its sensitive area A is perpendicular to the z-axis. And we then have a source of EM wave which sends out plane waves along the z-axis towards the receiver. What is the power of the EM wave received by the detector?

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \Rightarrow |\vec{S}| = \frac{1}{\mu_0} |\vec{E}| |\vec{B}| \sin \theta_{\vec{E}\vec{B}} = \frac{1}{\mu_0} |\vec{E}| |\vec{B}|$$

90°

But recall: $S = \frac{P}{A} \equiv I$ (Intensity)

Intensity: একক ক্ষেত্রফলে কত শক্তি যাচ্ছে।
 P = Power
 A = Area



And for plane waves: $\vec{E} = \vec{E}_0 \sin(kz - \omega t)$ $\vec{B} = \vec{B}_0 \sin(kz - \omega t)$

So: $\Rightarrow I = \frac{1}{\mu_0} E_0 \sin(kz - \omega t) B_0 \sin(kz - \omega t) = \frac{1}{\mu_0} E_0 B_0 \sin^2(kz - \omega t) \Rightarrow I = \frac{1}{\mu_0 c} E_0^2 \sin^2(kz - \omega t)$

সমীকরণ থেকে পাওয়া যায়:

Since: $E_0 = cB_0$

Important Features from EW Wave Intensity

$$I = \frac{1}{\mu_0 c} E_0^2 \sin^2(kz - \omega t)$$

$$I \propto E_0^2$$

Intensity amplitude-এর বর্গের সমানুপাতিক।

1. Intensity is proportional to E_0^2 , i.e., intensity is proportional to the square of the amplitude of the wave
যদি Amplitude দ্বিগুণ হয়:
 $I = (2)^2 = 4$, অর্থাৎ Intensity চারগুণ হবে।
2. Intensity is time-dependent! EM wave intensity fluctuates very rapidly with time. E.g., visible light has a frequency $\sim 10^{15}$ oscillations per second!

Intensity সময়ের সাথে পরিবর্তিত হয়।
কারণ: $\sin^2(kz - \omega t)$

EM Spectrum

ক্রম:

সব ধরনের EM Wave এখানে দেখানো হয়েছে

Gamma Ray

↓

X-Ray

↓

UV

↓

Visible Light

↓

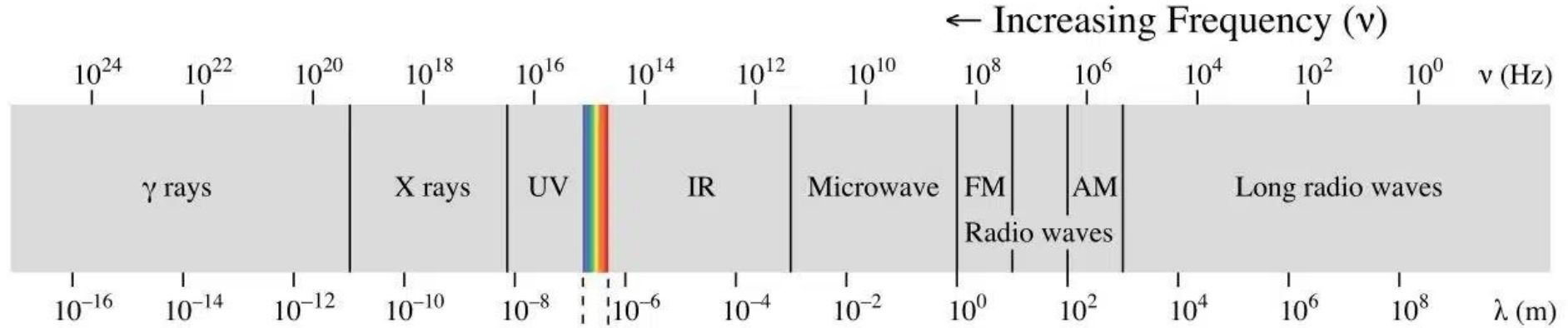
Infrared

↓

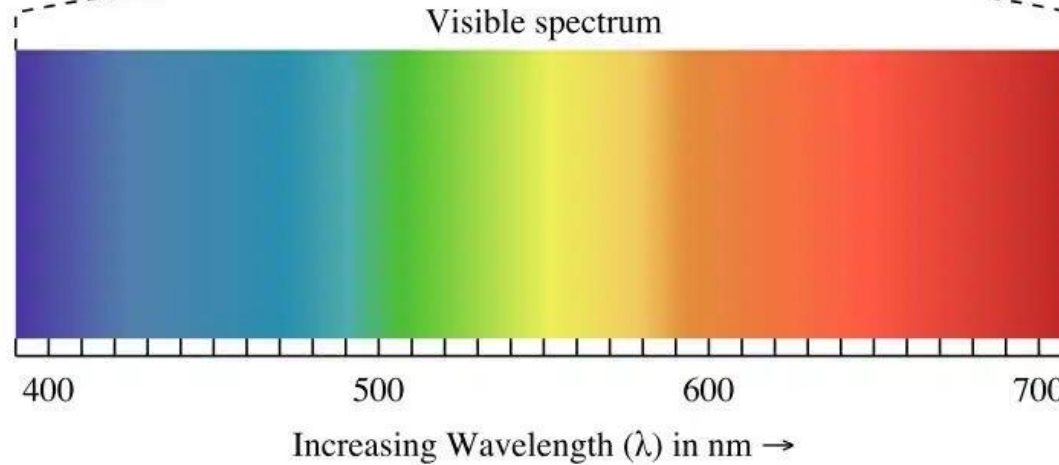
Microwave

↓

Radio Wave



Increasing Wavelength (λ) →



Frequency বেশি

-Gamma Ray

Wavelength বেশি

-Radio Wave

Visible Region

400nm-700nm

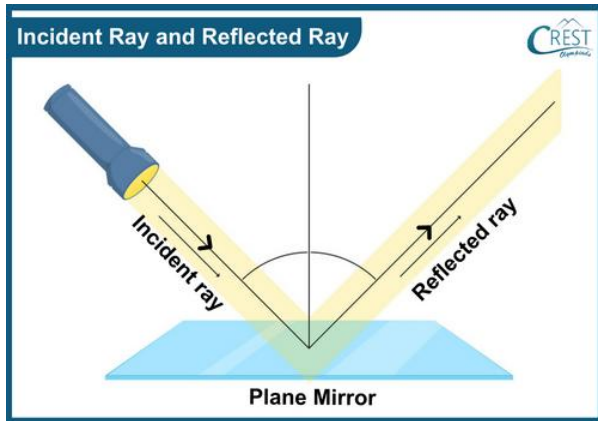
Why is light a wave?

কারণ আলো Wave-এর সব বৈশিষ্ট্য দেখায়।

- Because light demonstrates wave properties like

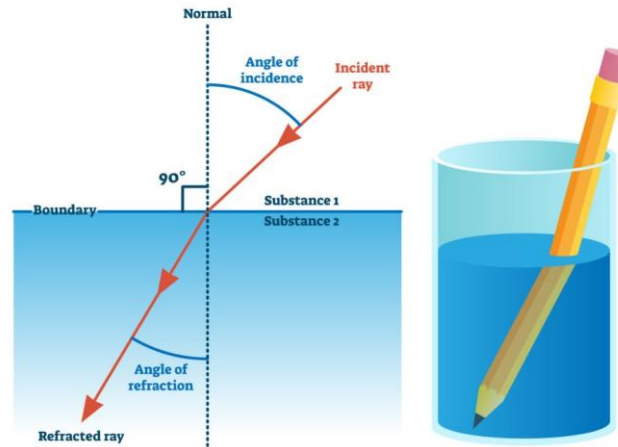
আয়নায় প্রতিফলন।

✓ Reflection



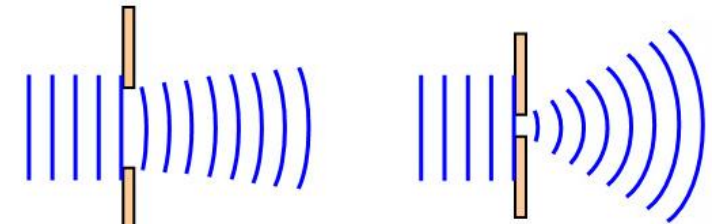
পানিতে পেন্সিল বাঁকা দেখায়।

✓ Refraction



ছোট ফাঁক দিয়ে গেলে তরঙ্গ ছড়িয়ে যায়।

✓ Diffraction



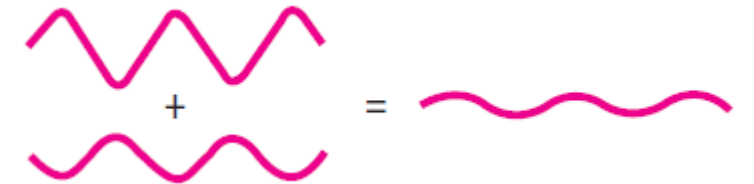
✓ Superposition



Constructive interference

উজ্জ্বল অংশ।

দুই তরঙ্গ মিলিত হলে।



Destructive interference

অন্ধকার অংশ।

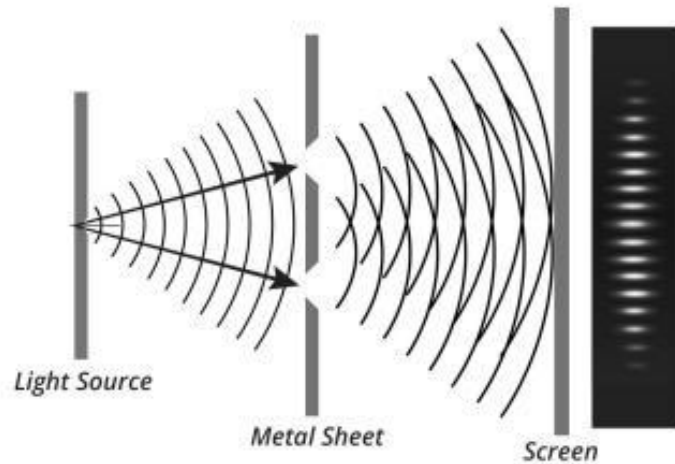
Young's Double Slit Experiment

এই পরীক্ষাই প্রথম প্রমাণ দেয়: Light behaves as a wave.

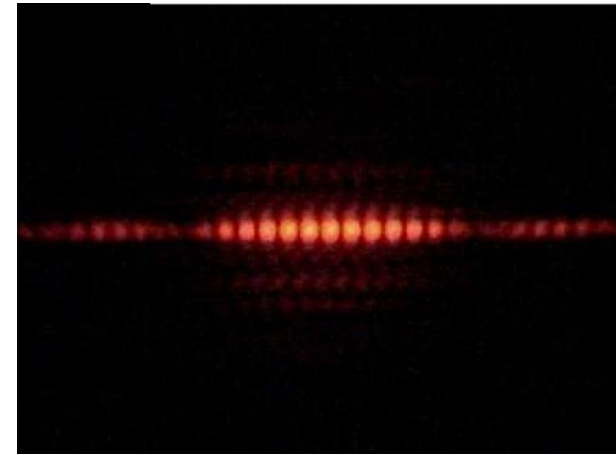


Thomas Young first demonstrated diffraction and interference of light waves, in 1801

- Experimental setup: shine a *monochromatic* light source through a pair of slits



Setup
Monochromatic Light
Two Slits
Screen



দুই Slit থেকে বের হওয়া আলো একে অপরের সাথে Interfere করে।

- Waves emerging from each slit spread out and interfere with each other.
- On a screen, a pattern of light and dark is seen showing evidence of superposition of light waves emerging from the slits

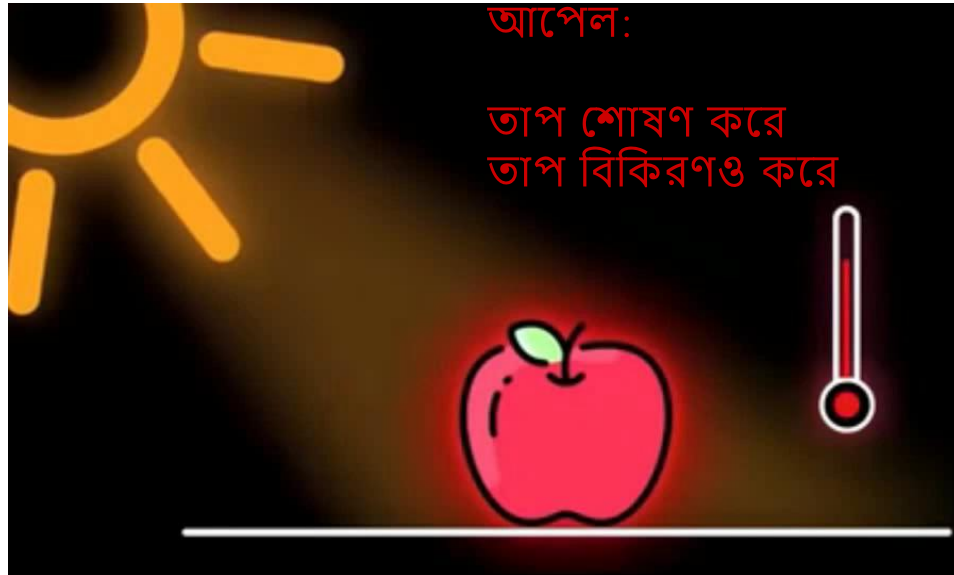
- At path lengths between the screen and the slits of $\lambda, 2\lambda, 3\lambda, \dots$ we see **bright fringes**

যখন $\Delta L = n\lambda$ তখন Constructive Interference হয়।

- At path lengths between the screen and the slits of $\frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \dots$ we see **dark fringes**

Emission and Absorption of EM radiation

- Consider an apple kept at room temperature and left there for some time....



- ✓ Apple absorbs thermal radiation, its temperature rises

It doesn't completely disintegrate. Why?

- ✓ Because it also *radiates* or *emits* thermal or EM radiation
- ✓ At a certain temperature, absorption and emission balances out. This is **thermal equilibrium**

একসময় Absorption = Emission হয়ে যায়। এটাকে বলে Thermal Equilibrium

- What causes the apple to radiate thermal energy?
কেন বিকিরণ করে? কারণ উত্তপ্ত Electron নড়াচড়া করে EM Radiation উৎপন্ন করে।
- ✓ Electrons gain energy through heat absorption and jiggle around giving off EM radiation
- ✓ Giving off EM radiation implies that the apple itself must be *glowing*
- ✓ But can you see an apple in the dark? Why not? EM radiation it emits is in IR region

Heating Metals

Room Temperature দেখতে স্বাভাবিক।



Metal at room temperature



High Temperature লাল হয়ে যায়।

Metal at high temperature

Very High Temperature সাদা/উজ্জ্বল হয়ে যায়।



Metal at extremely high temperature

কারণ: Temperature বাড়লে নির্গত EM Radiation-এর Frequency বাড়ে।

✓ Thus, heated metals emit EM radiation in varying wavelengths depending on temperature

✓ Emission radiation is over a *continuous spectra* – all wavelengths are emitted

যে আদর্শ বস্তু: সব Radiation শোষণ করে সব Radiation নির্গত করতে পারে তাকে Blackbody বলে।

✓ Ideally, an object that radiates all wavelengths should also absorb all wavelengths – this idealized object is called a **blackbody**